

Evidence-Calibrated Enterprise RAG for Hallucination-Resistant Question Answering

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Abstract—Retrieval-augmented generation (RAG) reduces but does not remove unsupported answers in enterprise question answering, where evidence may be incomplete, contradictory, or outdated. We study Evidence-Calibrated Enterprise RAG (ECERAG), a selective answering framework that scores the sufficiency of retrieved evidence from four signals—relevance, coverage, agreement, and source trust/recency—and answers, asks for clarification, or abstains accordingly. We evaluate it on a 60-question benchmark built from SEC filings, the NIST AI Risk Management Framework, the OWASP Top 10, and current/obsolete IETF specifications, under clean retrieval and three perturbations (irrelevant, contradictory, and outdated evidence), with five open-weight generators from 0.5B to 14B parameters, repeated cross-validation, paired bootstrap intervals, and two independent graders. Relative to a RAG baseline that must always answer, ECERAG cuts the error rate on answered questions by roughly two-thirds (e.g., 0.43 to 0.13 at 7B; 0.40 to 0.11 at 14B). A controlled decomposition shows that most of this gain comes from the generator’s own evidence-grounded refusal; the external sufficiency gate alone yields small reductions (at most 0.01–0.11), significant only for the 7B model or under outdated evidence. The sufficiency score is nevertheless a reliable ranking signal: among answered questions it orders wrong answers below right ones better than chance at every generator size from 1.5B to 14B. We conclude that *retrieve–assess–decide* is effective, that assessment is best led by grounded self-refusal, and that evidence-side scores are a complementary confidence signal rather than a standalone defense.

Index Terms—retrieval-augmented generation, hallucination, selective prediction, calibration, enterprise question answering, trustworthy AI

I. INTRODUCTION

Enterprise deployments of large language models (LLMs) need factual reliability and auditable behavior under uncertainty: in financial, legal, and engineering settings a confident wrong answer can cause material harm. Retrieval-augmented generation (RAG) grounds answers in retrieved documents [1], [2], but a standard pipeline still answers unconditionally, even when the retrieved passages are irrelevant, contradictory, or superseded.

ECERAG makes answering conditional on an explicit assessment of the evidence. A sufficiency score $S(q, E)$ routes each query to *Answer*, *Clarify*, or *Abstain*, in the tradition of selective prediction [3], [4]. Our research question is: *can evidence-sufficiency signals reduce unsupported answers in enterprise RAG without materially degrading coverage, and how much of any reduction is due to the evidence gate itself?*

A pilot of this design with 40 questions and a 0.5B-parameter generator found no reduction in selective risk, and attributed most errors to the generator. This paper replaces that pilot with a controlled study and makes four contributions:

- 1) A fully specified sufficiency score and selective policy (Sec. III), including a conflict check that detects near-duplicate passages disagreeing on a figure.
- 2) A generator-scale study (0.5B–14B) with repeated cross-validation, so every question is a held-out test item, and paired bootstrap confidence intervals (Sec. IV).
- 3) A decomposition that separates the evidence gate from the generator’s own refusal by adding a truly unconditional baseline and a gate-only system (Sec. V-B).
- 4) Evidence that the sufficiency score is a reliable *ranking* signal for answer correctness across scales, even where thresholded gating adds little (Sec. V-C).

II. RELATED WORK

Retrieval-augmented generation. RAG couples a parametric generator with non-parametric retrieval [1], building on dense retrieval [5], late interaction [6], and fusion-in-decoder readers [7]; surveys document persistent failures under weak or noisy retrieval [2]. **Adaptive and corrective retrieval.** Self-RAG [8] trains the generator to retrieve and critique on demand; Corrective RAG [9] detects poor retrieval and re-retrieves. Neither defines an explicit abstention policy calibrated to enterprise risk. **Selective prediction.** A selective model trades coverage for lower risk on the inputs it accepts [3], [10]; Kamath et al. [4] apply this to QA under domain shift, and LLMs have been shown to partly know what they know [11]—a point our decomposition makes concrete for RAG. **Evaluation.** RAG evaluation has moved toward faithfulness and context relevance [12], robustness to retrieval shift [13], and attribution [14]; LLM-as-judge grading [15] is increasingly used and should be cross-checked, as we do here.

III. METHOD

A. Retrieval

Documents are split into 220-word passages with a 40-word overlap. For a query q , BM25 [16] scores and cosine similarities of all-MiniLM-L6-v2 embeddings [17] are each min-max normalized over the corpus and averaged with equal weight. The top 20 passages are reranked with the

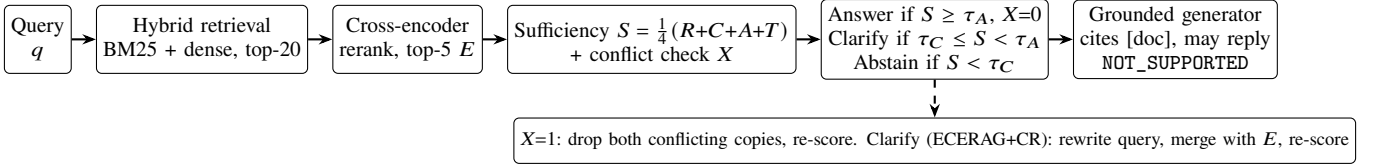


Fig. 1. ECERAG pipeline. Every component is specified in Sec. III.

ms-marco-MiniLM-L-6-v2 cross-encoder [18], and the top five form the evidence set $E = \{e_1, \dots, e_5\}$.

B. Evidence Sufficiency Score

$S(q, E) = \frac{1}{4}(R + C + A + T) \in [0, 1]$, with equal weights (fitting the weights did not help at this data size; Sec. V-E).

Relevance R is the mean cross-encoder probability $\sigma(\ell_i)$ over E , where ℓ_i is the reranker logit for (q, e_i) .

Coverage C is the fraction of the query’s salient terms (lower-cased alphanumeric tokens of at least three characters, excluding stop words) that occur in the concatenated evidence.

Agreement A is the mean over passage pairs (i, j) of the embedding cosine similarity s_{ij} , multiplied by 0.1 when the pair conflicts. Let N_i be the set of numbers in e_i . A pair conflicts if it is on the same topic ($s_{ij} > 0.55$) and $N_i \cap N_j = \emptyset$, or if both passages come from the same document, are near-duplicates ($s_{ij} > 0.9$), and $N_i \neq N_j$.

Trust/recency T is the mean over E of $0.7 b_d + 0.3 \max(0, 1 - \min(a_d, 20)/20)$, where $b_d = 0.35$ if the source document d is marked obsolete (superseded by a newer version in the corpus) and 1 otherwise, and a_d is the document’s age in years at a fixed reference date.

Conflict flag $X = 1$ if any same-document near-duplicate pair disagrees on a figure. A document cannot contradict itself, so $X = 1$ blocks a direct Answer: ECERAG drops both conflicting passages, re-scores the remainder, and answers only if it clears τ_A .

C. Selective Policy and Generation

ECERAG answers if $S \geq \tau_A$ and $X = 0$, asks for clarification if $\tau_C \leq S < \tau_A$, and abstains otherwise. The generator is prompted to answer only from the evidence, cite passages by document tag, and reply NOT_SUPPORTED when the evidence lacks the answer (*grounded self-refusal*). For year-over-year financial questions, a deterministic calculator supplies the computed difference, since small generators made arithmetic errors on correctly retrieved figures. The corrective variant ECERAG+CR handles the Clarify band: the generator rewrites the query, the new top-5 is merged with E , conflicts are removed, the merged set is reranked against the original question, and the result is accepted only if its score clears τ_A and exceeds the original S by at least 0.05.

IV. EXPERIMENTAL SETUP

A. Corpus and Benchmark

The corpus has six public documents (1,207 passages): Apple’s Form 10-K for fiscal years 2022 and 2023 (a current/superseded pair), the NIST AI RMF 1.0, the OWASP

Top 10:2021, and RFC 2616 and RFC 9110 (obsolete/current HTTP specifications). We hand-authored 60 questions verified against the sources: 15 single-hop, 15 multi-hop (most combining two document versions), 14 temporal/version-sensitive, and 16 unanswerable (false premise, out of corpus, or over-specific). Each answerable item has a gold answer and gold passage identifiers.

B. Perturbations

Each condition modifies the clean top-5 evidence of a question. *Irrelevant*: the lowest-ranked passage is replaced by the highest-scoring passage from a different document (55 applicable questions). *Contradictory*: the lowest-ranked slot is replaced by a copy of a gold passage with one number altered by $\pm 37\%$ (25 applicable: questions whose gold passage was retrieved and contains a number). *Counterfactual*: a passage from a current document is replaced by its nearest neighbor (embedding similarity) in the superseded version, i.e., what a stale index would return (46 applicable).

C. Systems

Unconditional RAG answers every question from E with the same prompt minus the refusal instruction. *Self-refusal RAG* uses the full prompt and declines only via NOT_SUPPORTED; this was the baseline of the pilot. *Evidence gate only* applies the S/X policy to the unconditional answers, isolating the sufficiency gate. *ECERAG* combines the gate with self-refusal and conflict handling; *ECERAG+CR* adds corrective retrieval. All systems share retrieval, reranking, evidence, and decoding, so they differ only in the decision step. Generators are Qwen2.5-Instruct models [19] with 0.5B, 1.5B, 3B, 7B, and 14B parameters, decoded greedily; the 14B model runs in 4-bit NF4 [20].

D. Calibration and Statistics

Thresholds are chosen inside training folds of a stratified 5-fold cross-validation repeated 20 times, so every question is scored out-of-fold. Candidate τ_A values are the observed training scores; we minimize $\text{risk} + 0.5(1 - \text{coverage})$ subject to $\text{coverage} \geq 0.25$, then choose τ_C on the ECERAG+CR outcome. Differences are reported with 95% paired bootstrap intervals (2,000 resamples of questions [21]), averaging over the 20 repeats within each resample.

E. Metrics and Grading

Coverage is the fraction of questions answered; *selective risk* is the error rate among answered questions, where any answer to an unanswerable question is an error; *abstention*

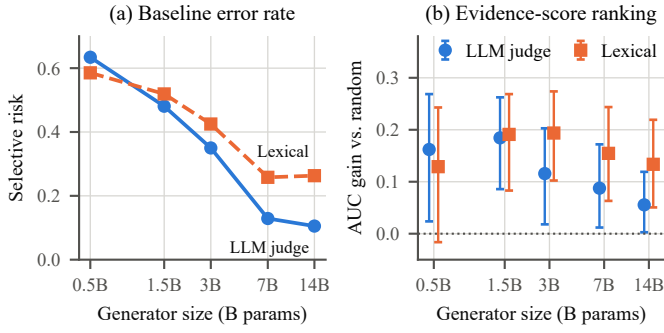


Fig. 2. (a) Selective risk of self-refusal RAG on clean evidence. (b) Gain of the sufficiency score’s risk-coverage AUC over a random ranking, on the questions the generator answered; bars are 95% bootstrap intervals.

recall is the fraction of unanswerable questions declined. The *risk-coverage AUC* [10] summarizes how well a score ranks answers (lower is better; a random ranking scores the overall error rate). Correctness is graded twice: (i) *lexically*, by recall of gold figures (with unit normalization, so that “\$383.3 billion” matches “\$383,285 million”) and keywords; and (ii) by an *LLM judge*, Phi-3.5-mini-instruct [22], chosen from a different model family than the generators to avoid self-preference [15]. The graders agree on 75–82% of answers; the judge more often credits correct answers phrased differently from the gold text. We report the judge as primary and state where the lexical grader differs.

V. RESULTS

A. Generator Scale

The generator is the dominant source of error at small scale. Self-refusal RAG’s selective risk falls from 0.63 at 0.5B to 0.11 at 14B (judge; 0.59 to 0.26 lexical) on clean evidence (Fig. 2a), although the retrieved evidence contains the answer for 80% of questions at every scale. From 3B upward the generator declines all 16 unanswerable questions by itself; it also declines answerable ones (13 of 44 at 7B, 6 of 44 at 14B), over half of them multi-hop questions.

B. Decomposing the Gain

Table I compares the four decision rules at 7B and 14B, and Table II gives the paired differences from the unconditional baseline. ECERAG reduces selective risk substantially and significantly on clean, irrelevant, and outdated evidence, for both generators and both graders—for example from 0.43 to 0.13 at 7B and from 0.40 to 0.11 at 14B on clean evidence.

The decomposition shows where this gain comes from. Self-refusal alone reaches almost the same operating point: 0.13 at 7B and 0.11 at 14B; at 14B, ECERAG and self-refusal make identical decisions. The evidence gate alone moves the unconditional system only slightly. Its reductions are significant under the judge for the 7B model on clean and outdated evidence and for the 14B model on outdated evidence (Table II); none is significant under the lexical grader. The gate also catches few unanswerable questions (abstention

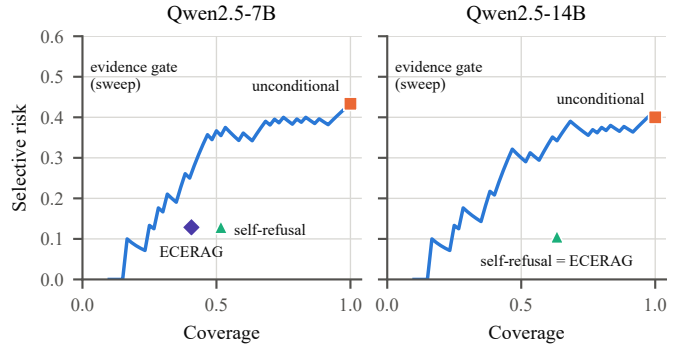


Fig. 3. Risk-coverage on clean evidence (LLM judge). The curve sweeps the evidence-gate threshold over unconditional answers; markers are cross-validated operating points.

recall 0.09–0.33 versus 1.00 for self-refusal), because false-premise questions retrieve evidence that looks relevant. Fig. 3 shows the full trade-off: sweeping the gate threshold over unconditional answers traces a curve that lies well above the self-refusal operating point at every coverage.

C. The Sufficiency Score as a Ranking Signal

Although thresholding S adds little to self-refusal, S ranks answers well. Restricted to the questions the generator answered (so no credit comes from its own refusals), the risk-coverage AUC of S beats a random ranking at every scale from 1.5B to 14B under both graders, with bootstrap intervals that exclude zero, except for the 14B judge interval, whose lower bound is 0.00 (Fig. 2b). At 7B, S reaches an AUC of 0.04 against 0.13 for a random ranking (judge; 0.10 against 0.26 lexical). Table III breaks the score down: relevance and coverage carry most of the signal, trust/recency helps, and agreement is weakest on its own.

D. Contradictory Evidence and Corrective Retrieval

The conflict check flagged all 25 contradictory evidence sets and 1 of 60 clean ones. The check alone was the decisive fix: the pilot’s agreement rule required two passages to share *no* numbers, so it could never detect a copy with one figure changed, and all systems behaved identically under this perturbation. Acting on the flag is harder. Refusing every flagged case yields zero risk at zero coverage. Dropping both conflicting copies recovers 38% (7B) and 28% (14B) coverage, but at a selective risk of 0.27 and 0.14 (judge), not significantly different from the unconditional baseline’s 0.20 and 0.12, because the dropped pair usually contains the passage with the answer. Self-refusal RAG remains strong on this condition (0.14 and 0.05 risk at 88% coverage). Corrective retrieval that replaces the evidence wholesale increased risk at 14B (0.05 to 0.23 on contradictory evidence); the merged, margin-gated version left coverage and selective risk unchanged in every condition.

E. Ablations

Fitting the four weights by logistic regression, and adding an NLI-based grounding score of the drafted answer, did

TABLE I
SELECTIVE RISK / COVERAGE (%) BY DECISION RULE, CROSS-VALIDATED OVER 60 QUESTIONS. UPPER BLOCK: LLM JUDGE; LOWER BLOCK: LEXICAL GRADER. LOWER RISK IS BETTER.

Grader	System	Qwen2.5-7B			Qwen2.5-14B		
		Clean	Irrelevant	Outdated	Clean	Irrelevant	Outdated
Judge	Unconditional RAG	0.43 / 100	0.47 / 100	0.63 / 100	0.40 / 100	0.38 / 100	0.48 / 100
	Evidence gate only	0.38 / 83	0.45 / 86	0.60 / 88	0.39 / 91	0.38 / 92	0.46 / 92
	Self-refusal RAG	0.13 / 52	0.16 / 58	0.36 / 54	0.11 / 63	0.03 / 62	0.24 / 54
	ECERAG	0.13 / 41	0.14 / 41	0.34 / 43	0.11 / 63	0.03 / 62	0.24 / 54
Lexical	Unconditional RAG	0.52 / 100	0.51 / 100	0.57 / 100	0.47 / 100	0.40 / 100	0.46 / 100
	Evidence gate only	0.51 / 74	0.50 / 74	0.54 / 77	0.46 / 91	0.40 / 92	0.46 / 93
	Self-refusal RAG	0.26 / 52	0.25 / 58	0.40 / 54	0.26 / 63	0.21 / 62	0.24 / 54
	ECERAG	0.23 / 34	0.20 / 35	0.24 / 35	0.30 / 53	0.23 / 50	0.26 / 47

TABLE II
CHANGE IN SELECTIVE RISK RELATIVE TO UNCONDITIONAL RAG: 95% PAIRED BOOTSTRAP INTERVALS. INTERVALS EXCLUDING ZERO ARE IN BOLD.

Model	Evidence	Evidence gate only		ECERAG	
		Judge	Lexical	Judge	Lexical
7B	Clean	[-.11, -.01]	[-.05, .04]	[-.45, -.16]	[-.45, -.14]
7B	Outdated	[-.07, -.01]	[-.06, .02]	[-.47, -.11]	[-.50, -.16]
14B	Clean	[-.04, .02]	[-.04, .02]	[-.42, -.17]	[-.32, -.02]
14B	Outdated	[-.05, -.00]	[-.02, .03]	[-.40, -.09]	[-.36, -.06]

TABLE III
RISK-COVERAGE AUC OF EACH SIGNAL ON ANSWERED QUESTIONS (JUDGE / LEXICAL). LOWER IS BETTER; "RANDOM" IS THE EXPECTED AUC OF A RANDOM RANKING.

Score	1.5B	7B	14B
Random	.48 / .52	.13 / .26	.11 / .26
<i>S</i> (equal weights)	.30 / .33	.04 / .10	.05 / .13
<i>R</i> relevance	.28 / .33	.04 / .12	.05 / .15
<i>C</i> coverage	.26 / .29	.05 / .09	.05 / .14
<i>A</i> agreement	.45 / .53	.09 / .25	.05 / .25
<i>T</i> trust/recency	.39 / .39	.05 / .11	.05 / .16

not significantly improve selective risk over equal weights at 7B (judge, 95% interval of the difference from the baseline $[-0.10, +0.03]$). The grounding score’s apparent AUC advantage disappeared once the generator’s own refusals were excluded from the ranking. Removing any single signal changed clean-condition risk by at most 0.03. With the pilot’s fixed-split calibration (17 calibration and 43 test questions), the grid selected $\tau_A = 0.80$ for both 7B and 14B, giving zero errors at 26% coverage against 0.13 and 0.10 for the baseline: a conservative operating point that the cross-validated cost function does not choose.

VI. DISCUSSION

Implications. For enterprise RAG, the most effective single control in our study is to instruct the generator to answer only from cited evidence and to refuse otherwise, provided the generator is large enough to follow the instruction (3B and up here). Evidence-sufficiency scores add a calibrated, inspectable confidence signal on top of that. They are useful for ranking, triage, and routing outdated or conflicting evidence to review,

and they provide an audit trail that self-refusal alone does not. Whether to gate on them is a risk-tolerance choice, which Fig. 3 makes explicit.

Limitations. The benchmark has 60 questions from six documents, so many intervals are wide; results may not transfer to other domains without recalibration. Correctness is graded automatically; the two graders agree on 75–82% of answers, and we have not yet validated either against human judgment (an annotation sheet of 100 stratified answers is released with the code). Unanswerable questions were authored by us and may be easier to detect through self-refusal than those arising in practice. All generators come from one model family, and the 14B model was quantized. The perturbations are synthetic; in particular, the contradiction check targets near-duplicate passages that disagree on a figure, which matches our injection procedure but is only one kind of real-world conflict.

VII. CONCLUSION

Enterprise RAG should move from retrieve-then-generate to retrieve-assess-decide. In a controlled study across five generator sizes, ECERAG reduced the error rate on answered questions by roughly two-thirds relative to unconditional RAG. Most of the reduction comes from grounded self-refusal by the generator; the evidence-sufficiency gate adds a small, sometimes significant improvement, and its score reliably ranks answer correctness from 1.5B to 14B parameters. Future work will validate grading against human annotation, enlarge the benchmark, and use the sufficiency score to drive targeted retrieval rather than blanket abstention.

CODE AND DATA AVAILABILITY

The pipeline, corpus sources, benchmark, raw model outputs, and analysis scripts are available at <https://github.com/yomisys/LLM>.

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